
ZERU SHI

CONTACT INFORMATION

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RESEARCH INTERESTS

- **LLM/VLM Reasoning and Post-training:** I am highly interested in post-training for LLMs and VLMs. My focus is on developing more effective reinforcement learning algorithms, as well as leveraging simple and interpretable methods, to improve reasoning capabilities through post-training.
- **Agentic AI:** I am also interested in Agentic AI, especially in the design of agent memory. I am particularly drawn to a storage centric perspective, where memory is treated as a first-class component. My goal is to build principled memory systems that can significantly enhance agents' capabilities in complex reasoning and real world problem solving.

EDUCATION

Rutgers University, New Brunswick, USA Ph.D. in Computer Science Department of Computer Science	<i>September 2025 - May 2030(Expected)</i> Advisor: Dr. Ruixiang Tang
Dalian University of Technology, Liaoning, China Bachelor of Science in Software Engineering Department of Software Engineering	<i>September 2021 - June 2025</i> Advisor: Dr. Risheng Liu

INTERNSHIP EXPERIENCE

Research Scientist Intern, NVIDIA	<i>June 2026 - August 2026</i>
Research Scientist Intern, Shanghai AI Laboratory	<i>November 2024 - April 2025</i>

SELECTED PUBLICATIONS

* denotes an equal contribution.

[ICML' 26] **Zeru Shi**, Zhenting Wang, Fan Yang, Qifan Wang, Ruixiang Tang. "A Single Layer to Explain Them All: Understanding Massive Values in Large Language Models" In *The International Conference on Machine Learning*

[ICLR' 25] **Zeru Shi***, Kai Mei*, Mingyu Jin, Yongye Su, Chaoji Zuo, et al. "From Commands to Prompts: LLM-based Semantic File System" In *The International Conference on Learning Representations*

[IEEE TCSVT] **Zeru Shi**, Zengxi Zhang, Kemeng Cui, Ruizhe An, Jinyuan Liu, Zhiying Jiang. "Se-FENet: Robust Deep Homography Estimation via Semantic-Driven Feature Enhancement" In *IEEE Transactions on Circuits and Systems for Video Technology (IF = 11.1)*

[EMNLP' 25] Yongye Su, Yucheng Zhang, **Zeru Shi**, Bruno Ribeiro, Elisa Bertino "Castle: Causal Cascade Updates in Relational Databases with Large Language Models" In *Empirical Methods in Natural Language Processing (Main)*

[Under Review] **Zeru Shi**, Yingjia Wan, Zhenting Wang, Qifan Wang, Fan Yang, Elisa Kreiss, Ruixiang Tang. "Meaningless Tokens, Meaningful Gains: How Activation Shifts Enhance LLM Reasoning". In submission.

[Under Review] **Zeru Shi***, Minghao Guo*, Qingyue Jiao*, Yihao Quan, et al. “MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory”. In submission.

[Under Review] **Zeru Shi***, Kai Mei*, Yihao Quan, Dimitris N Metaxas, Ruixiang Tang. “Improving Visual Reasoning with Iterative Evidence Refinement”. In submission.

[Under Review] **Zeru Shi**, Zhenting Wang, Yongye Su, Weidi Luo, Hang Gao, et al. “Auto-Prompt Generation is Not Robust: Prompt Optimization Driven by Pseudo Gradient”. In submission.

SELECTED HONORS AND AWARDS

- Dalian excellent undergraduate graduates, China
- National Scholarship, China
- Special scholarship of NOK Corporation, Japan

MISCS

Programming Skills and Tools: Python(Strong), C/C++(Proficient), MySQL(Proficient)

Language: English, Chinese, Japanese